

COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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In this paper the authors look at EM algorithm based estimation of parameters of lognormal, Weibull, gamma and generalised gamma distributions when the survival data is subject to left truncation and right censoring. Extensive simulations are done to study the performance of point and interval estimators of the parameters of interest. Model discrimination issues are also discussed in the last chapter. The paper is essentially based on Balakrishnan and Mitra (2011, 2012, 2013a, 2013b).

Let T , C , and V denote the lifetime, censoring time and truncation time of a unit, respectively. Let F , G and H , respectively denote their distribution functions. Suppose $P(C > V) = 1$ and both C and V are independent of T . Under left truncation and right censoring we do not observe failures if $T < V$. If $T \geq V$, then we observe (X, δ, V) where $X = \min(T, C)$ and $\delta = I(T \leq C)$. Let $a_F = \inf\{t : F(t) > 0\}$ and $b_F = \inf\{t : F(t) = 1\}$. a_G, a_H, b_G, b_H are defined analogously. Assume $a_H \leq \min(a_F, a_G)$ and $b_H \leq \min(b_F, b_G)$. Let $a > a_H$. Given n i.i.d observations on (X, δ, V) , a nonparametric maximum likelihood estimator of $F(t|a) = P(T \leq t | T > a)$ is given by (Wang, 1987)

$$\hat{F}_n(x|a) = 1 - \prod_{a \leq u \leq x} \left[1 - \frac{N_F(du)}{nR(u)} \right],$$

where $R(u) = \frac{1}{n} \sum_{i=1}^n I(V_i \leq u \leq X_i)$ and $N_F(u) = \sum_{i=1}^{n_d} I(T_i \leq u)$ and $N_F(du) = N_F(u) - N_F(u-)$. Note $nR(T_i) > N_F(dT_i)$ for $i = 1, \dots, n_d$.

The nonparametric estimator $\hat{F}_n(x|a)$ has also been studied by Keiding and Gill (1990) and Lai and Ying (1991). Gross and Lai (1996), Sundaram (2009) and Shen (2009a) have considered the case when the truncation distribution H involves covariates.

If the distribution of V is parametrized as $H(x, \theta)$ and those of T and C are not specified, then $\hat{F}_n(x|a)$ is no longer the M.L.E. Shen (2009b) proposed consistent semiparametric estimators for $F(t|a)$. The case $H(x, \theta) = \frac{x}{\theta}$ (the length biased case) is discussed in detail.

A brief mention of our work on parametric estimation based on EM algorithm for bivariate survival data is given.

Sometimes the life data of interest is bivariate in nature. Two common examples are study on lifelength of Danish twins by Hougaard, Harvald and Holm (1992) and paired data on patients of colon cancer where the data consists of the time from treatment to recurrence of the cancer and the

time from treatment to death (Lin, Sun and Ying, 1999). Paired data could consist of blindness in the left/right eye, failure time of the left/right kidney or age at death of parent/child in a genetic study. However, one or both components of the paired data could be subject to random censoring. It could arise because a parent and /or the child might be alive till the end of the study and hence the failure time would be censored.

Nandi and Dewan (2010) considered the bivariate Marshall-Olkin Weibull distribution where both components of the random vector are subject to independent random right censoring. Dewan and Nandi (2013) considered the bivariate generalised exponential distribution where the two components are left censored. The two censoring schemes are naturally interpretable for the two models. In both the papers we have used the EM algorithm (conditional) to find maximum likelihood estimators of the parameters of interest. The algorithm performs well (in terms of mean square error) for sample sizes 50, 75 and 100. The results are good even in the presence of censoring.

Nonparametric estimators of the bivariate survival function when both components are subject to left truncation and right censoring have been discussed by Shen and Yan (2008) and Shen (2010).

For statistical analysis of competing risks data under left truncation and right censoring one could refer to Sankaran and Antony (2007) and Shen (2011, 2012).

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